

JEE Mains 2019 Chapter wise Question Bank

Heights and Distances – Questions

Q1

Consider a triangular plot ABC with sides $AB = 7$ m, $BC = 5$ m and $CA = 6$ m. A vertical lamp-post at the mid point D of AC subtends an angle 30° at B. The height (in m) of the lamp-post is:

- (1) $\frac{3}{2}\sqrt{21}$ (2) $\frac{2}{3}\sqrt{21}$
(3) $2\sqrt{21}$ (4) $7\sqrt{3}$

10 Jan Morning

Q2

If the angle of elevation of a cloud from a point P which is 25 m above a lake be 30° and the angle of depression of reflection of the cloud in the lake from P be 60° , then the height of the cloud (in meters) from the surface of the lake is:

- (1) 60 (2) 50
(3) 45 (4) 42

12 Jan Evening

Q3

Two vertical poles of heights, 20 m and 80 m stand apart on a horizontal plane. The height (in meters) of the point of intersection of the lines joining the top of each pole to the foot of the other, from this horizontal plane is :

- (1) 15 (2) 18 (3) 12 (4) 16

8 April Evening

Q4

Two poles standing on a horizontal ground are of heights 5m and 10m respectively. The line joining their tops makes an angle of 15° with the ground. Then the distance (in m) between the poles, is:

- (1) $5(2 + \sqrt{3})$ (2) $5(\sqrt{3} + 1)$
(3) $\frac{5}{2}(2 + \sqrt{3})$ (4) $10(\sqrt{3} - 1)$

9 April Evening

Q5

ABC is a triangular park with $AB = AC = 100$ metres. A vertical tower is situated at the mid-point of BC. If the angles of elevation of the top of the tower at A and B are $\cot^{-1}(3\sqrt{2})$ and $\operatorname{cosec}^{-1}(2\sqrt{2})$ respectively, then the height of the tower (in metres) is :

- (1) $\frac{100}{3\sqrt{3}}$ (2) $10\sqrt{5}$ (3) 20 (4) 25

10 April Morning

Q6

A 2 m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/sec., then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is:

- (1) $25\sqrt{3}$ (2) $\frac{25}{\sqrt{3}}$ (3) $\frac{25}{3}$ (4) 25

12 April Morning

Q7

The angle of elevation of the top of a vertical tower standing on a horizontal plane is observed to be 45° from a point A on the plane. Let B be the point 30 m vertically above the point A. If the angle of elevation of the top of the tower from B be 30° , then the distance (in m) of the foot of the tower from the point A is:

- (1) $15(3 + \sqrt{3})$ (2) $15(5 - \sqrt{3})$
(3) $15(3 - \sqrt{3})$ (4) $15(1 + \sqrt{3})$

12 April Evening



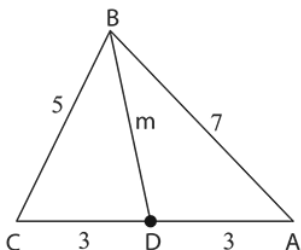
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Heights and Distances - Solutions

Q1

(2) Let the height of the lamp-post is h .



By Appolonius Theorem,

$$2\left(BD^2 + \left(\frac{AC}{2}\right)^2\right) = BC^2 + AB^2$$

$$\Rightarrow 2(m^2 + 3^2) = 25 + 49 \Rightarrow m = 2\sqrt{7}$$

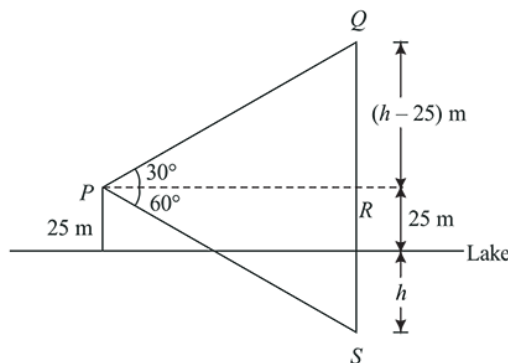
$$\tan 30^\circ = \frac{h}{BD}$$

$$\Rightarrow h = 2\sqrt{7} \times \frac{1}{\sqrt{3}} = \frac{2\sqrt{21}}{3}$$

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Q2

(2) Let height of the cloud from the surface of the lake be h meters.



$\therefore$

In $\triangle PRQ$:

$$\tan 30^\circ = \frac{h - 25}{PR}$$

$\therefore$

$$PR = (h - 25) \sqrt{3} \quad \dots(i)$$

and in $\triangle PRS$: $\tan 60^\circ = \frac{h + 25}{PR}$

$$PR = \frac{h + 25}{\sqrt{3}} \quad \dots(ii)$$

Then, from eq. (i) and (ii),

$$(h - 25) \sqrt{3} = \frac{h + 25}{\sqrt{3}}$$

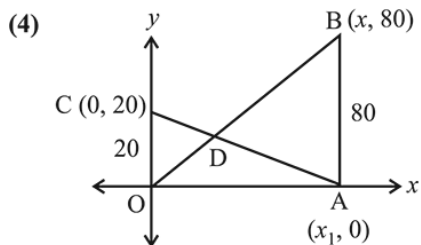
$\therefore$

$$h = 50 \text{ m}$$

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Q3

Heights and Distances



Equations of lines OB and AC are respectively

$$y = \frac{80}{x_1}x \quad \dots(i)$$

$$\frac{x}{x_1} + \frac{y}{20} = 1 \quad \dots(ii)$$

$\therefore$ equations (i) and (ii) intersect each other

$\therefore$ substitute the value of x from equation (i) to equation (ii), we get

$$\frac{y}{80} + \frac{y}{20} = 1$$

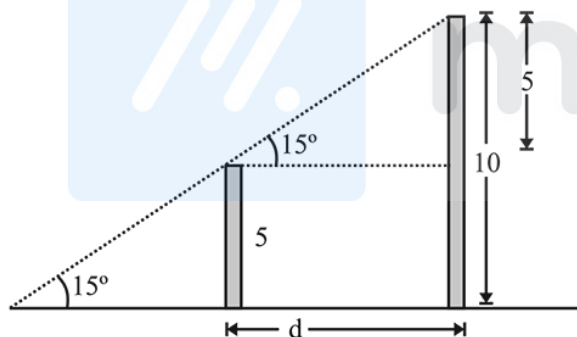
$$\Rightarrow y + 4y = 80 \Rightarrow y = 16 \text{ m}$$

Hence, height of intersection point is 16 m.

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Q4

(1)



By the diagram,

$$\tan 15^\circ = \frac{5}{d} \Rightarrow d = \frac{5}{\tan 15^\circ} = \frac{5(\sqrt{3}+1)}{\sqrt{3}-1}$$

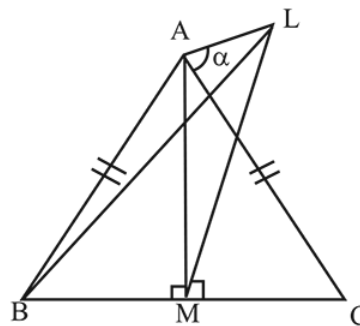
$$= \frac{5(4+2\sqrt{3})}{2} = 5(2+\sqrt{3})$$

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Q5

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(3) Let the height of the vertical tower situated at the mid point of BC be h .



In $\triangle ALM$,

$$\cot A = \frac{AM}{LM}$$

$$\Rightarrow 3\sqrt{2} = \frac{AM}{h} \Rightarrow AM = 3\sqrt{2}h$$

In $\triangle BLM$,

$$\cot B = \frac{BM}{LM} \Rightarrow \sqrt{7} = \frac{BM}{h} \Rightarrow BM = \sqrt{7}h$$

In $\triangle ABM$ by Pythagoras theorem

$$AM^2 + MB^2 = AB^2$$

$$\therefore AM^2 + MB^2 = (100)^2$$

$$\Rightarrow 18h^2 + 7h^2 = 100 \times 100$$

$$\Rightarrow h^2 = 4 \times 100 \Rightarrow h = 20$$

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Q6

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(2) According to the question,

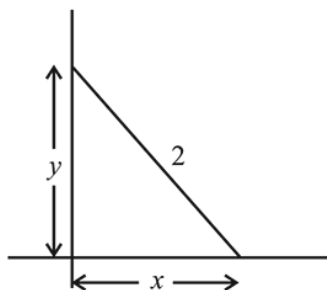
$$\frac{dy}{dt} = -25 \text{ at } y = 1$$

By Pythagoras theorem, $x^2 + y^2 = 4$... (i)

When $y = 1 \Rightarrow x = \sqrt{3}$

Diff. equation (i) w. r. t. t ,

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$



$$\Rightarrow x \frac{dx}{dt} + y \frac{dy}{dt} = 0 \Rightarrow \sqrt{3} \frac{dx}{dt} + (-25) = 0$$

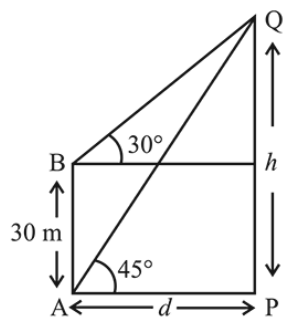
$$\Rightarrow \frac{dx}{dt} = \frac{25}{\sqrt{3}} \text{ cm/s}$$

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Q7

(1) Let the height of the tower be h and distance of the foot of the tower from the point A is d .

By the diagram,



$$\tan 45^\circ = \frac{h}{d} = 1$$

$$h = d \quad \dots (i)$$

$$\tan 30^\circ = \frac{h-30}{d}$$

$$\sqrt{3}(h-30) = d \quad \dots (ii)$$

Put the value of h from (i) to (ii),

$$\sqrt{3}d = d + 30\sqrt{3}$$

$$d = \frac{30\sqrt{3}}{\sqrt{3}-1} = 15\sqrt{3}(\sqrt{3}+1) = 15(3+\sqrt{3})$$